

Germany's Efficient Homes

by Nicole Pillen and Nana Doerrie

As energy prices on the world market rise and the debate on how to address climate change intensifies, energy-efficient home retrofits have become a vital issue in Germany. The residential building sector consumes the largest proportion of site energy used in Germany, and the lion's share—about 75%—of this use is attributable to space heating. Consequently this share has to be reduced radically for overall energy consumption and carbon dioxide (CO₂) emissions to decline. Existing buildings are of particular importance. At present, older buildings use on average 225 kilowatt-hours per square meter (kWh/m²) (71,325 British thermal units per square foot (Btu/ft²)) annually—or 3 times the energy required to heat a new building. Yet despite rising energy prices, retrofit projects are still not fully exploiting potential energy savings. Currently, over half of the potential savings that are economically feasible remain untapped after a renovation.

To realize these potential savings, the German energy agency, dena, has initiated the Efficient Homes project. Launched in 2003, this project uses best practices to

- develop, test, and implement cost-effective low-energy standards for retrofits that are superior to the federal requirements for new buildings;
- publicize and improve innovative technologies for increasing the energy efficiency of existing buildings; and
- educate and encourage builders, designers, architects, and the general public to implement all

economically feasible techniques for energy efficient retrofits of existing buildings.

Many educational efforts and promotional events have been held to publicize the project. These include workshops for architects, planners, and the housing industry, as well as special events for consumers. Remodeling guidelines have been developed for each of these different target groups. In addition, regional competence networks, which consist primarily of local energy agencies, have been created to increase awareness of efficient retrofitting techniques. The Federal Promotional Bank (KfW Förderbank) has also been doing its part to support project participants by offering low-interest loans and partial payments to reduce the costs of borrowing to finance these retrofits.

These efforts have paid off quite well. Since 2003 more than 140 buildings comprising 2,230 residential units have undergone energy-efficient retrofits (see Figure 1). The project includes every type of residential building—from single-family homes to an apartment building with 295 units, so renovation recommendations and best-practice case studies now exist for almost every residential building type. The buildings represent a mix of commercially and privately owned residences; 49% belong to housing cooperatives and commercial landlords, and 51% belong to private individuals. Of the latter, 24% are private landlords and 27% are owners of single-family and two-family homes.



JOSEPH BUCKEL



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This house in Feuchtwangen, Bavaria, pictured during (top) and before (bottom) the retrofit, is the oldest building in the Efficient Homes project. It's 344 years old, to be exact.

In many cases, these retrofits employed highly innovative passive-house components, which, until recently, have been used almost exclusively in new buildings. A passive house is an ultra-low energy house that is built according to a rigorous energy standard: the building must not use more than 15 kWh/m² per year (4,750 Btu/ft² annually) in heating energy. This is usually achieved by employing superinsulation materials and special windows that combine triple-pane insulation glazing with air seals and specially developed thermally broken window frames, which reduce heat loss through the frame.

To appreciate the accomplishments of the Efficient Homes project, one must first understand the differences between German and U.S. building stock. Most buildings in Germany are not built out of wood; 86% of single-family and two-family homes are built out of concrete, concrete block, or brick, while only 14% are stick built. In



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Innovative solutions can be used to meet the very low energy standards of the Efficient Homes project.

National Distribution of the 143 "Efficient Homes" Participants

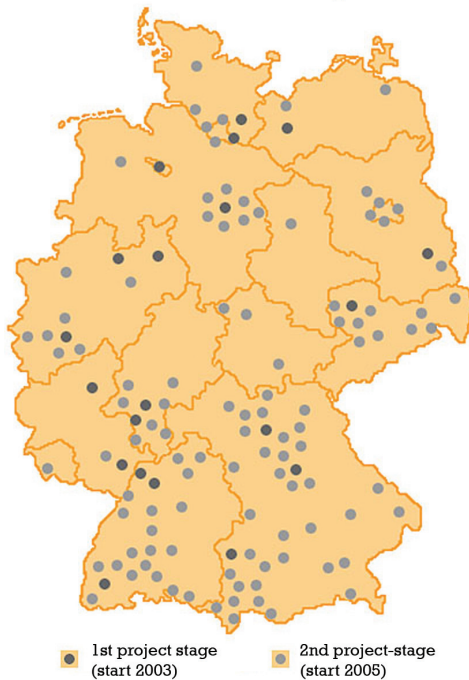


Figure 1. Since 2003 more than 140 buildings comprising 2,230 residential units have undergone energy-efficient retrofits.

the United States, 95% of single-family homes are stick built.

Two-thirds of the residential buildings in Germany were built before 1978. The average rate for new developments has been 1% per year for

the past several years. This means that existing buildings dominate Germany's building stock. Approximately 35% of Germany's residential buildings must be completely retrofitted now or within the next two decades, with another 28% of them needing to be completely rebuilt for the second time sometime in the next 20 years. Altogether, this means that roughly 50% of Germany's older buildings will be retrofitted within the next 20 years. It is therefore essential to implement cost-effective low-energy retrofitting standards immediately. Otherwise the energy demand for space heating and domestic hot water cannot be significantly reduced.

Tightening Energy Standards

Until 2006 participants in the Efficient Homes project could choose to meet either of two energy standards—the Efficiency Standard or the Future Standard—both of which are related to the energy requirements for new buildings. The Efficiency Standard requires the energy consumption of a retrofitted building to be 30% less than that of a comparable new building. To meet the Future Standard, the energy consumption of a retrofitted building must be 50% less than that of a comparable new building. Starting in

2007, all participants in the Efficient Homes project have been obligated to meet the Future Standard. (For both standards, "energy consumption" refers to the primary energy demand of the building, which is defined as the energy embodied in natural resources—coal, crude oil, sunlight, uranium—that has not undergone any human-caused conversion or transformation. In the United States, this energy is referred to as source energy.)

These standards do not specify particular materials or technical installations that must be used. This is done to allow for innovative solutions to meet the standard. However, a ventilation system is required in every building. The project has already been such a success that the Efficiency Standard has been incorporated into the Federal Promotional Bank's regular offers for energy-saving retrofits that are not part of the Efficient Homes project.

The Efficient Homes retrofit projects have proved that it is possible to reduce the energy consumption of old buildings significantly. These energy savings were achieved using only standard construction techniques. Before the retrofit, the average energy demand of these 140 buildings was 336 kWh/m² (106,512 Btu/ft²) annually. After the retrofit, it was reduced to an average of 44 kWh/m² (13,948 Btu/ft²) annually. This is far below the minimum requirement for comparable new buildings of 98 kWh/m² per year (31,066 Btu/ft² annually). These results clearly demonstrate that energy-efficient methods of construction can reduce the energy demand of an existing building by an average of over 80% (see Figure 2).

The Oldest Low-Energy Building

The oldest building in the Efficient Homes project is quite old—344 years old, to be exact. This timber-framed building was built in the small village of Feuchtwangen, Bavaria, in 1663, just 15 years after the end of the Thirty Years' War. It is a two-story single-family home. The retrofit was completed in summer

2007. The owner, Josef Buckel, said that he wanted to update the building, and he wanted to save energy and protect the environment.

What was done to modernize this house? The most important measure was the installation of superinsulation made of polystyrene to significantly reduce heat leakage through the walls, roof, and floor. A 4-inch (10-centimeter) thermal insulation composite system, with a U-value of 0.31 Watts per square meter times degrees Kelvin ($\text{W/m}^2\text{K}$), was installed on the facade. The insulation on the roof is now 8 inches (20 centimeters) thick, with a U-value of 0.13 $\text{W/m}^2\text{K}$. It was extremely important to identify and eliminate the building's thermal bridges in order to achieve the best possible results. The casement windows were retrofitted with energy-efficient glazing. The energy for space heating and the provision of warm water is now supplied by three different sources: a geothermal pump, a solar-thermal collector, and a mechanical heat recovery ventilation system. Because the new insulation made the building more airtight, it was necessary to install a mechanical ventilation system to prevent condensation and mold.

The results of the retrofit are impressive. The building's total primary energy supply (TPES) was reduced by 85%. Before the retrofit, the TPES was 572 kWh/m^2 annually (181,324 Btu/ft^2 annually). It is now just 36.5 kWh/m^2 annually (11,570 Btu/ft^2 annually), based on modeling of the newly retrofitted building. (These modeled savings will be checked against the actual energy consumption of the building. All project participants are required to report this data for the first two years following modernization.) Mr. Buckel can be proud; by retrofitting his home, he has made an enormous contribution to the overall reduction of greenhouse gases. According to dena's calculations, his house now saves 39 metric tons of CO_2 per year! And he profits directly, too, because the final or site energy—which he pays for—adds up to only 12 kWh/m^2 per

year (3,804 Btu/ft^2 annually). Although the building is more than 300 years old, it is now more efficient than most new buildings in Germany, and Mr. Buckel needn't worry about rising energy prices.

The Most Efficient Single-Family Home

The project's most efficient single-family home is a two-story, 2,551 square foot (237 square meter) timber building, with an addition built from concrete. It was built 50 years ago on the banks of the beautiful Lake Bodensee, one of Europe's biggest lakes in the south of Germany. When the house was originally built in the 1950s, no one was paying attention to energy efficiency. When it came time to retrofit, the current owner, Andreas Koch, wanted to reduce the building's extremely high annual energy demand of 255 kWh/m^2 annually (80,835 Btu/ft^2 annually). Koch is an architect

specializing in energy-efficient construction who works for the European Institute for Energy Research. His objective was to preserve the building's typically '50s style while significantly reducing its energy demand. He was extremely careful in choosing the measures he wanted to apply, so the building would still resemble the family house that he and his mother and sister had inherited from his grandparents.

What was done? Again insulation played a major role. A thermal insulation composite system made of polystyrene with a thickness of 12 inches (30 centimeters) and a U-value of 0.102 $\text{W/m}^2\text{K}$ was installed on the timber construction part of the building. New wooden cladding was



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The most important measure during the Feuchtwangen retrofit was the installation of superinsulation made of polystyrene to significantly reduce heat leakage through the walls, roof, and floor.

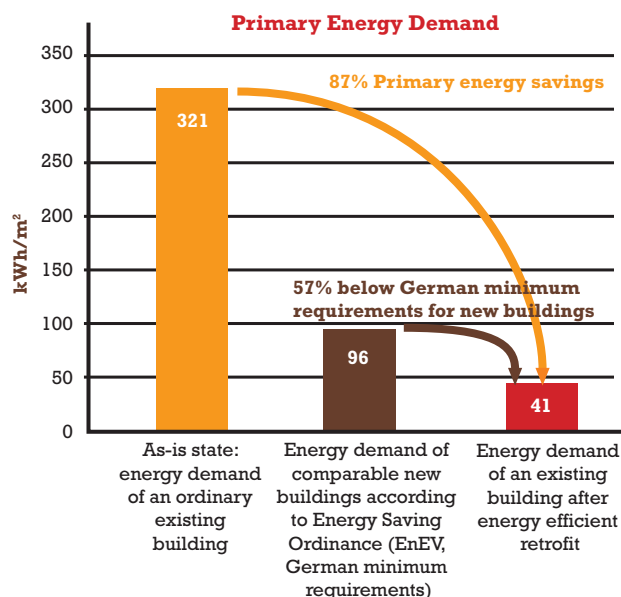


Figure 2. After the retrofits, the energy demand of the 140 buildings was reduced to an average of 44 kWh/m^2 (13,948 Btu/ft^2) annually. This is far below the minimum requirement for comparable new buildings of 98 kWh/m^2 per year (31,066 Btu/ft^2 annually).

installed over the polystyrene. A thermal insulation composite system was installed on the concrete addition as well, with the roof and basement getting 12 inches (30 centimeters) of polystyrene. The side of the house facing the lake underwent the most dramatic changes. The balcony was torn down and the windows were replaced with larger, highly efficient, three-pane thermal units.

The result is remarkable: a beautiful view of the lake and the mountains; brighter, more spacious rooms; and



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Koch, the owner of the house pictured above, is an architect specializing in energy-efficient construction who works for the European Institute for Energy Research. His objective was to preserve his home's typically 1950s style (bottom) while significantly reducing its energy demand (top).

outstanding energy efficiency. You wouldn't know that you were standing in front of a window if you were standing blindfolded just 12 inches away—proof of the windows' superb thermal quality. The building is now extremely airtight, so much so that to ensure good air quality, a mechanical ventilation system had to be installed. But Mr. Koch is not wasting energy with his ventilation system. He chose a system that recovers 80% of the heat that would normally be lost while exhausting the indoor air.

The heating system was completely retrofitted as well. The old-fashioned oil heating was replaced by an innovative geothermal heat pump that uses brine and water to produce 7.8 kilowatts,

enough to provide a constant supply/ system temperature of 95°F (35°C). A 101 square foot (9.4 square meter) solar-thermal collector was also installed on the roof of Mr. Koch's house. It is one of the new, efficient vacuum tube units, with an output of 3.75 kilowatts. The energy provided by the geothermal heat pump and the solar collector warm

the water in a storage tank that is then distributed throughout the house for heating and domestic hot water. The solar collector supplies 51% of the energy needed to heat the domestic hot water, and 8% of the energy needed to heat the house.

To complete the energy-efficient heating system retrofit and to improve the residents' comfort, Mr. Koch installed an underfloor radiant heating system. Instead of wasting energy by heating a room at ceiling height as some forced-air systems do, underfloor heating warms the air at floor level, and that warm air then rises through the living space, warming its occupants and losing heat as it rises. Very little

energy is therefore wasted, and Mr. Koch's rooms are heated in a way that makes them more comfortable—warm feet and a cool head, rather than cold feet and a hot head.

The result of all these changes is that the annual TPES of 255 kWh/m² annually (80,835 Btu/ft² per yr) before the retrofit was reduced by 86% to just 34 kWh/m² annually (10,778 Btu/ft² annually) after refurbishment. That's a third of the regularized minimum requirement for a comparable new building, which would be 107 kWh/m² annually. Like Mr. Buckel, Mr. Koch has become almost independent of rising energy prices. The building's final or site energy demand is now calculated to be as little as 11 kWh/m² annually (3,487 Btu/ft² annually). And Mr. Koch has done more than his share to protect the environment and the climate. With the improvements to the building, his home now emits 30 metric tons of CO₂ less per year.

Success All Around

The Efficient Homes project has been a great success. By the middle of 2007, more than 1,300 owners had applied for the program and are now committed to highly energy-efficient retrofits. The two examples described above—the Buckel and Koch homes—show that these renovations are not miracles. Standard measures and technologies that are readily available in the marketplace were used to attain these exemplary energy standards.

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For more information:

For more information on these projects, visit www.dena.de and www.zukunft-haus.info or contact doerrie@dena.de.